

The simulation pipeline

Every simulator run is a pipeline: compile sources, elaborate the design hierarchy, then execute time

Verilator's compile-to-binary path

- Verilator's common flow is compile with the C++ emit flag
- Elaboration happens inside Verilator's front end
- Status READY in the literacy lab means every stage completed

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Compile → elaborate → run

INTERACTIVE TOOL

Compile → elaborate → run

Document-aid literacy: the simulation toolchain stages **compile** → **elaborate** → **run** (iverilog / Verilator story). Starter: all three pass — pipeline READY.

Starter example: compile → elaborate → run all **pass** — pipeline READY.

Load starter example

Challenges 0 / 22 cleared

Quiz: **compile**: Compile typically means...

☐ parse / check sources and emit an intermediate form

☐ only open GTKWave

☐ skip hierarchy binding

☐ place-and-route

Show hint

Check

Next

Idle

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Core ideas

compile

Parse sources → IR / C++ emit.

elaborate

Bind hierarchy, params, generates.

run

vvp scheduler or C++ binary.

READY

All three pass — full pipeline.

Lab

SCENARIO

all pass

Load preset

Set stage

Scan pipeline

Demo gap

Explain

Reset

Lab starter

learn_verilator — sim pipeline

Real Verilator practice

- In Track A, open EXAMPLES and run one minimal Verilator compile that emits C++, then make
- Say out loud which log lines belong to compile versus link versus run
- If the binary exits immediately, check whether the testbench host actually stepped time

Pitfalls to watch

- Do not treat “verilator ran” as “simulation succeeded”, did you build and run the host?
- Do not debug waves when compile never finished
- Do not compare Icarus and Verilator commands flag-for-flag; compare stages
- And remember READY is a whole pipeline, not one green checkbox

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, reach ready on the pipeline diagram
- In Track A, run compile, build, and execute once on a tiny design
- When you are ready, take the short quiz, then continue to Verilator lint